Shashank Lahoti

615-877-0985 | slahoti7@gatech.edu | Atlanta, GA | linkedin.com/in/shashank-lahoti | github.com/Shashank1775

Education

Georgia Institute of Technology Atlanta, GA

B.S. Computer Science (Intelligence, Systems & Architecture) | *B.S. Mathematics* | *GPA: 4.0/4.0* *May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Design & Analysis of Algorithms, Computer Organization & Programming, Objects & Design, Discrete Mathematics, Linear Algebra, Multivariable Calculus

Experience

AI & Engineering Summer Intern May 2025 – Jul. 2025

Deloitte – Product & Engineering *Hybrid – Nashville, TN*

- Cut estimated M&A deal execution time ∼10–20% in demos by prototyping a multi-agent AI system in Python + LangChain
- Automated first-pass due diligence by building Legal, Financial, and Technical agents on Azure OpenAI that audit data rooms
- Grounded agent outputs in deal documents via LangChain orchestration with RAG over embeddings in vector databases
- Supported cross-border compliance checks (GDPR, antitrust, SEC) by encoding regulatory rules in a Python rule engine
- Improved successive demos by folding mentor code-review critique into agent prompts, LLM evals, and orchestration logic

Machine Learning Intern Oct. 2024 – Apr. 2025

Cadi (liquidation reseller: overstock via Facebook Marketplace & eBay) *Nashville, TN*

- Replaced spreadsheet-driven warehouse workflows by shipping a custom CRM and inventory platform with a 3-person team
- Enabled real-time tracking of 800+ goods by location, interest, value, and supplier by designing the PostgreSQL schema
- Streamlined live customer Q&A by prototyping an AI sales assistant (OpenAI-API chatbot + generated video)
- Raised customer conversion ∼18% and cut average wait time ∼42% in an ∼80-customer pilot by deploying the assistant
- Shipped features in weekly sprints with a 3-person team, reworking designs on code-review and customer feedback

Bioinformatics & ML Intern May 2023 – May 2025

ChemTex Intl. *Remote*

- Kept a 2+ year research program on track by leading a 5-person team, mentoring juniors, and liaising with the PI
- Automated structure-based screening pipelines in Python (AutoDock Vina, MD, MM/GBSA); 2nd author, 2 manuscripts
- Named inventor on patent filing 202541052143 A (June 2025) by co-developing the invention with 6+ researchers

Projects

RideWise – Smart School Bus Tracking | *Next.js, Node.js, MongoDB, Azure, Google Maps, Arduino* 2024

- Served bus data through 9 input-validated, Jest-tested REST endpoints by building the Node.js backend on MongoDB
- Scoped district admins to their own data by implementing JWT + bcrypt auth with server-side role checks
- Captured boarding and location events by building the on-bus Arduino unit (GPS + RFID scans)
- Showed parents live per-bus journeys and sent alert emails via a GPS ingestion service and Azure Communication Services
- Won the 2024 Congressional App Challenge and pitched the MVP to the district's Director of Transportation for adoption

Livo – Wearable Overdose Detection | *React Native, TypeScript, Express, MongoDB, Arduino* 2023 – Present

- Captured physiological signals by writing Arduino firmware (GSR, IR temp, IMU) and a Python serial-to-HTTP bridge
- Delivered live charts and BLE pairing by building an Express + MongoDB API (20 routes, JWT auth) and a React Native app
- Reached ∼87% detection accuracy across ∼13 subjects in science-fair validation by training a TensorFlow classifier

Hypothesis-First Prompting – LLM Reasoning Study | *Python, Azure OpenAI, Jupyter* 2025

- Tested whether stating a hypothesis before reasoning improves LLM accuracy by running a controlled study on GPT-3.5/4o
- Logged ∼24k responses across 6 prompt variants × 8 benchmarks; preprocessed data and built tables in Pandas/NumPy
- Lifted GPT-3.5's baseline from 79.5% to 82.9% with planning-first prompts via a reproducible harness (retry, JSON logs)

Technical Skills

Languages: Python, Java, C/C++, TypeScript, JavaScript, SQL (PostgreSQL), R, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, Express, React Native (Expo), TensorFlow, scikit-learn, Pandas, NumPy
Backend & Cloud: REST APIs, MongoDB/Mongoose, JWT & bcrypt, NextAuth/OAuth, AWS, Google Cloud (GCP), Azure (App Service, Static Web Apps, ACS, OpenAI), Google Maps API
Tools & Practices: Git/GitHub, GitHub Actions CI, Linux, Jest, ESLint, Postman, code review, embedded C (Arduino)
AI & Agents: LangChain, multi-agent orchestration, RAG, vector databases, OpenAI APIs, LLM evals, Claude Code

Honors & Awards

USAMO Qualifier (1 of ∼230); **USAJMO Qualifier** (1 of ∼200); **2x AIME Distinguished Honor Roll** (top 1%)
Congressional App Challenge Winner; **National STEM Festival Champion**; **Summer Science Program International Fellow**